

Intermediates

Intermediates are thought to be the "halfway" species that should exist if one group of organism has evolved from another. In 1861, discovery of the earliest fossil bird, *Archaeopteryx*, provided important support for this theory. The fossil clearly combines the reptile characteristics of a dinosaur in its skeleton, but at the same time has the uniquely bird-like feature of feathers.



Lungfish

Living intermediates

The lungfish can breathe oxygen directly from air and has paired fleshy fins for swimming. Lungfish arose about 380 million years ago when animals first stepped on land; it is thought that the first land-going tetrapods (four-footed animals) evolved from lungfish-like intermediates.

Human impact

For centuries humans have had an impact on animal and plant habitats. For example, some animals have had to adapt to new environments created by human settlement and industry. Also, by selective breeding, scientists have altered the genetic make-up of some plants and animals to create superior crops and meat for human consumption. Fruits, such as grapes and oranges, have been bred so that they no longer have seeds and are therefore easier to digest.

Tomatoes are bred to have a longer shelf life.



Seedless orange



Seedless grape

Artificial selection

Scientists can manipulate the genetic make-up of plants and animals by artificially selecting particular strains according to need. For instance, genetically altered products, such as tomatoes with enhanced flavour, are already on sale. Dog breeders, too, can alter the temperament, shape, size, and colour of dogs. The ability to alter genetic make-up raises many ethical questions.



Shar pei dog

Impact by pollution

It was discovered that during the Industrial Revolution, light coloured peppered moths, previously camouflaged from bird predators, became vulnerable on soot blackened trees. However, darker coloured peppered moths did survive and went on to breed more dark moths.



Peppered moths

Evidence

Much of the evidence for evolution is based on fossils. These show how life originated from simple forms in the sea and then evolved to occupy land, freshwater, and air. Study of the common characteristics of organisms suggest how they could be inter-related. Also, investigations into genetics and its molecular basis have led to a greater understanding of how evolution works.



Fossil of fern plant

Fossil evidence

Fossils of animals and plants show that there have been significant changes throughout geological time. Fossils show how the first land living plants appeared some 440 million years ago, but were not sufficiently well established to form forests until about 320 million years ago.

Molecular evidence

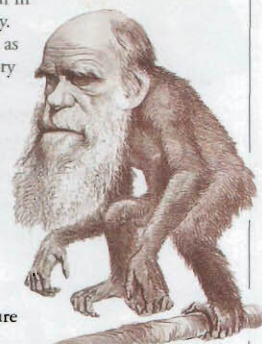
Often, if two new species evolve from a common ancestor, their DNA begins to change. For example, both the red and grey squirrel feed on acorns. However, the red squirrel lacks the digestive enzyme which breaks down chemicals in the acorn. This has led to a decline of red squirrels in the British Isles.



Grey squirrel population outnumbers that of the red squirrel in Britain

Creation theories

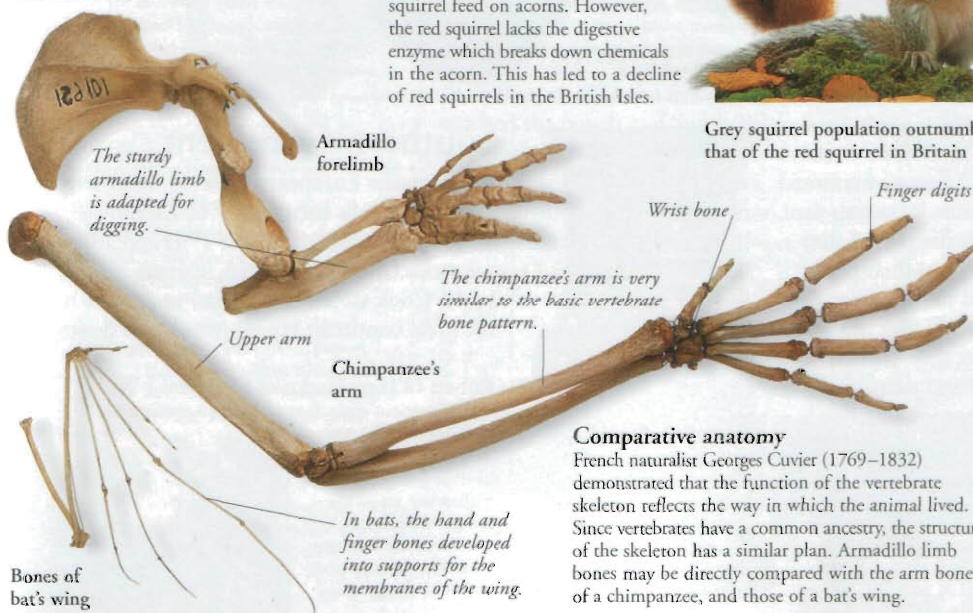
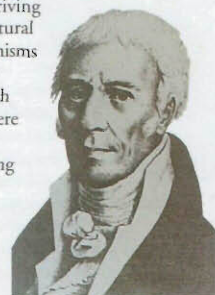
Evolutionists believe that life on Earth has evolved progressively over thousands of millions of years, originally from non-living materials. However, some people argue that all life forms on the Earth were created by design, in or close to the form in which they exist today. Although creationists as a group reject the theory of evolution, many accept that life forms on Earth can change. They believe that God created various kinds of creatures that have diversified within each kind until the present day.



19th-century caricature of Charles Darwin

Jean Baptiste de Lamarck

French naturalist Jean Baptiste de Lamarck (1744-1829) proposed a theory of evolution that broke the prevailing idea of the fixity of species. He claimed that by striving to fit into their natural surroundings, organisms changed in bodily form, and that such transformations were passed on to their offspring. According to his theory the giraffe developed a long neck by stretching to eat leaves from trees.



The sturdy armadillo limb is adapted for digging.

Armadillo forelimb

Upper arm

Chimpanzee's arm

Wrist bone

Finger digits

The chimpanzee's arm is very similar to the basic vertebrate bone pattern.

In bats, the hand and finger bones developed into supports for the membranes of the wing.

Bones of bat's wing

Comparative anatomy

French naturalist Georges Cuvier (1769-1832) demonstrated that the function of the vertebrate skeleton reflects the way in which the animal lived. Since vertebrates have a common ancestry, the structure of the skeleton has a similar plan. Armadillo limb bones may be directly compared with the arm bones of a chimpanzee, and those of a bat's wing.

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